Estimating Leaf Area Index and Leaf Nitrogen Content for Individual *Populus* Tree Using UAV Hyperspectral Data With Dual-Branch Deep Learning Architecture

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***Abstract*—Uncrewed aerial vehicles (UAVs) hyperspectral remote sensing provides a flexible approach for quantifying leaf area index (LAI) and leaf nitrogen content (LNC) of individual *Populus* tree. However, current methods neglect the influence of spatial scale, feature form, and model structure, resulting in unclear scale determination and limited spectral–spatial feature utilization. This study proposed a dual-branch deep learning (DL) architecture to fully exploit spatial and spectral features across multiple scales and feature forms. Hyperspectral image cubes of individual trees were generated at four scales. Three feature forms were evaluated, including all bands, selected bands, and vegetation indices (VIs). The dual-branch comprises parallel spectral and spatial branches, where spectral branch processes the averaged canopy reflectance, and the spatial branch uses the canopy image as input. The spectral branch integrates 1-D convolutional neural network (1-D CNN), attention mechanism (AM), and long short-term memory (LSTM) modules, while the spatial branch employs 2-D CNN, residual learning, and AM modules. In each branch, the optimal feature forms and spatial scales were determined and then used for developing the dual-branch models. The results showed that the optimal spatial resolutions for LAI and LNC estimation were 0.10 and 0.05 m, respectively. Both traits achieved optimal accuracy with selected bands used in the spectral branch and VIs used in the spatial branch. The proposed dual-branch models outperformed conventional DL models, with R^2 of 0.861 $\pm$ 0.006 for LAI and 0.842 $\pm$ 0.009 for LNC. This method provides an accurate high-throughput approach for monitoring of *Populus* germplasm, facilitating precision breeding and phenotyping applications.**

***Index Terms*—Deep learning (DL), leaf area index (LAI), leaf nitrogen content (LNC), Populus, uncrewed aerial vehicles (UAVs).**

I. Introduction

Populus is one of the typical fast-growing woody plants with advantages of large biomass and high economic value [1]. It is a valuable resource for the timber and paper industries and is also considered ideal for phytoremediation due to its ability to absorb pollutants like heavy metals [2]. However, *Populus* exhibits a high sensitivity to drought and salt-alkali stress, which presents significant challenges to its cultivation in the arid, saline, and sandy regions of northern China [3]. Therefore, selecting and breeding germplasm resources resistant to drought and soil salinization is crucial for improving the productivity of *Populus* and supporting both socio-economic development and environmental sustainability.

Leaf area index (LAI) and leaf nitrogen content (LNC) are critical phenotypic traits that indicate drought and salt-alkali tolerance in *Populus* species. Accurate estimation of these traits at the individual tree level is essential for high-throughput phenotyping and precision breeding, as it captures intrapopulation variability that is otherwise lost in plot- or stand-level averages. LAI is the total one-sided leaf area ratio to the ground area [4]. It is widely used to assess canopy structure and the photosynthetic potential of vegetation. Similarly, LNC reflects the nutrient status of plants, as nitrogen is a fundamental component of chlorophyll and proteins involved in photosynthesis [5]. Accurate and real-time monitoring of LNC provides valuable insights into the dynamic physiological response of plants to environmental stresses. Therefore, precise monitoring of LAI and LNC across different *Populus* genotypes is essential for advancing biological breeding [6]. Traditional methods for obtaining LAI and LNC data rely on manual and laboratory techniques. These methods are accurate but destructive and time-consuming, making them unsuitable for large-scale and high-throughput applications. In contrast, remote sensing technology provides a rapid, nondestructive, and cost-effective method for monitoring these

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traits over larger scales [7]. Optical satellite data observed by Landsat-7/8, Sentinel-2, and Gaofen-1/2 has been widely applied for LAI and LNC estimation in agricultural, forestry, and grassland ecosystems [8], [9]. However, the relatively low spatial resolution of satellite data limits its applicability at individual tree scales. In contrast, uncrewed aerial vehicles (UAVs) systems have offered another approach to obtaining remote sensing data. UAVs can be equipped with a variety of sensors, which enable flexible data acquisition at specific locations and times, independent of the constraints of satellite revisit cycles or atmospheric interference [10], [11]. These capabilities make UAVs remote sensing an ideal technology for LAI and LNC estimation at the individual tree level.

Variations in vegetation canopy reflectance properties are driven by changes in biophysical and biochemical parameters, which enable optical remote sensing to serve as an effective approach for LAI and LNC estimation. Various sensor systems, including RGB, multispectral, and hyperspectral platforms, have demonstrated capability in quantifying spectral variations across different wavelength ranges. Hyperspectral imaging systems are particularly advantageous, providing continuous spectral sampling of plant canopies with high spectral fidelity. This capability allows for precise discrimination of subtle vegetation characteristics [12], making UAV hyperspectral systems particularly promising for developing robust LAI and LNC estimation models. The accuracy of biophysical parameter estimation using UAV hyperspectral data is influenced by three factors, including spatial scale, form of input features, and model architecture. Spatial resolution represents a fundamental consideration, as it directly determines both data acquisition efficiency and the strength of the relationship between spectral information and target parameters [13], [14]. While finer spatial resolution provides enhanced within-pixel detail, excessively high resolution may introduce noise and nonvegetative signals that can degrade heterogeneity though mixed pixels, potentially increasing uncertainty in biophysical and biochemical parameter estimation [15]. Therefore, determining an optimal spatial resolution is critical to balance the tradeoff between capturing sufficient canopy details and minimizing noise or mixed-pixel effects in parameter estimation. Recently, increasing attention has been paid to the impact of UAV imagery spatial resolution on the estimation of crop phenotypic traits [16], [17]. However, most studies have focused on homogeneous crop canopies, while the influence of spatial scale on estimating the biophysical and biochemical parameters of individual trees is not yet well-understood.

Relevant and high-quality input features are essential for enabling models to extract meaningful patterns. In the estimation of LAI and LNC at the individual tree level, the selection and representation of input features are critical to model performance. Hyperspectral imagery, owing to its high spectral resolution, offers strong potential for LAI and LNC estimation by capturing subtle variations in canopy reflectance. However, using full-spectrum hyperspectral data may introduce redundant and irrelevant information, increasing the risk of overfitting and computational burden. Previous studies have primarily adopted two strategies to reduce redundancy, including selecting sensitive bands through feature selection algorithms (which is described as data-driven method) and employing narrowband vegetation indices (VIs) designed to enhance sensitivity to specific vegetation traits (which is described as physics-driven method) [18], [19], [20], [21], [22], [23]. However, selected bands and VIs represent two distinct feature representation strategies, yet their relative effectiveness remains uncertain. Recently, deep learning (DL) has provided an end-to-end solution for automatically extracting features from raw data, capturing complex nonlinear relationships. However, applying DL to hyperspectral data is challenged by high dimensionality. While dimensionality reduction through selected bands or VIs can mitigate computational costs, it remains unclear whether this simplification compromises key spectral information and affects the accuracy of DL models in LAI and LNC estimation.

Model architecture plays a crucial role in optimizing algorithms for the estimation of LAI and LNC in individual *Populus* trees. Utilizing the same input features with different model architectures often leads to different estimation results. Compared to statistical and traditional machine learning algorithms, DL offers greater flexibility in model design. This flexibility enables the development of DL models with various architectures to handle different data types and tasks effectively. For instance, convolutional neural networks (CNNs) excel at learning meaningful contextual features from imagery, while recurrent neural networks, such as long short-term memory (LSTM) networks, capture sequential dependencies, and temporal dynamics within the hyperspectral and multitemporal remote sensing data. However, single-model architectures often struggle to adequately capture both spatial and sequential dependencies. Recently, research has focused on hybrid architectures that combine two or more models to achieve superior performance. For instance, CNN–LSTM models have been successfully applied to crop yield estimation and leaf water content estimation [24], [25], [26]. In addition, hybrid dual-branch architectures also have been employed for LAI and LCC estimation in crops, where 1-D CNNs and 2-D CNNs are used to process spectral and spatial features, respectively [27]. Compared with CNN–LSTM models, dual-branch hybrid architectures explicitly separate spatial and spectral feature extraction, which reduces feature entanglement at early stages and enables more effective representation learning tailored to the characteristics of each modality. However, the flexibility of dual-branch architectures in existing studies remains limited. In particular, spectral and spatial features exhibit different sensitivities to spatial resolution and spectral band configurations, which are often not explicitly considered or jointly optimized in current dual-branch designs. Moreover, to date, the application of dual-branch DL architectures for estimating biophysical and biochemical parameters at the individual tree level has received limited attention, with most related studies still focusing on homogeneous crop canopies.

Therefore, this study aims to develop accurate LAI and LNC estimation models for individual *Populus* trees using double-branch hybrid DL architecture based on UAV hyperspectral data. Three sub-objectives are discussed: 1) identifying the sensitive bands for LAI and LNC estimation of individual *Populus* trees; 2) evaluating the effects of different spatial

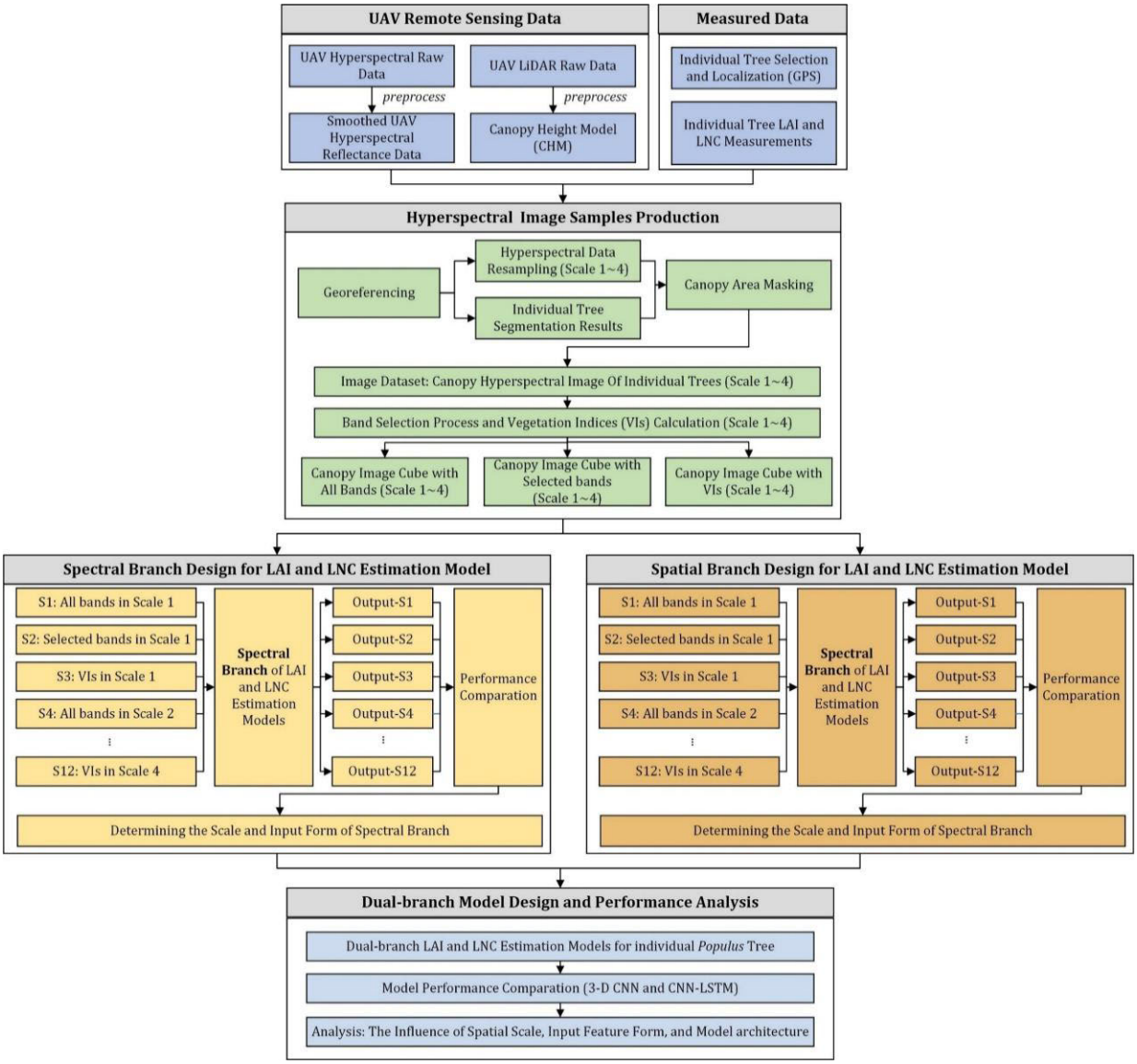


Fig. 1. Workflow of this study.

scales and input feature forms on LAI and LNC estimation in each branch; and 3) evaluate the estimation accuracy of the proposed method by comparing with different DL architectures.

II. Materials and Methods

A. Overview

Fig. 1 illustrates the overall workflow of this study. First, UAV-based raw remote sensing data (including hyperspectral imagery and LiDAR) and field measurements (such as GPS locations, LAI, and LNC of individual *Populus* trees) were collected. After preprocessing, smoothed hyperspectral imagery and a canopy height model (CHM) were generated and georeferenced. The hyperspectral imagery was then resampled into four spatial scales (Scale 1 to Scale 4), while the CHM data were used to perform individual tree segmentation. Image datasets at four spatial scales were constructed by masking the canopy area of each tree on the hyperspectral imagery. This study compared the effects of three input feature forms, including all bands, feature selected bands, and VIs. Accordingly, three types of canopy image cubes were generated at each scale. After data preprocessing and sample preparation, the inputs for the spectral and spatial branches of the LAI and LNC models were defined. Each branch first established its algorithm framework, and the optimal input form and scale were determined based on comparative results across different spatial scales and input forms. Subsequently, dual-branch LAI and LNC estimation models for individual *Populus* trees were developed, and their performance was compared with that of 3-D CNN (3-D CNN) and CNN-LSTM models. Finally, the effects of spatial scale, input feature form, and model architecture on model performance were discussed.

B. Study Area

The study area is in the National Superior *Populus* Seed Base in Shishou County, Hubei, China (Fig. 2). Shishou

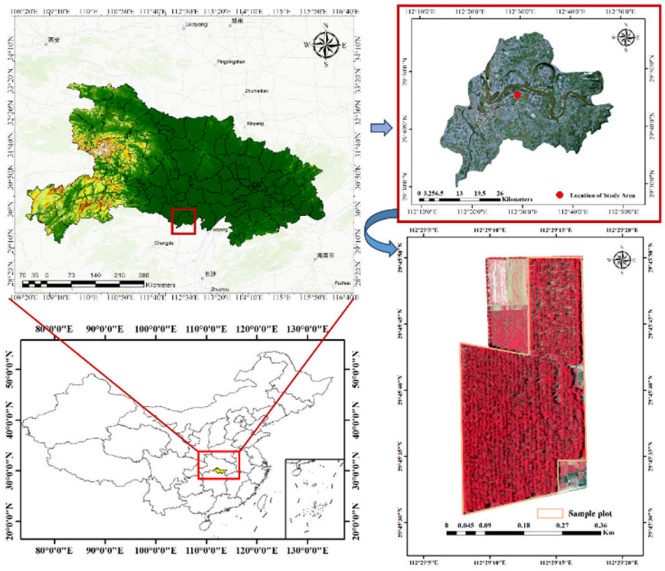


Fig. 2. Location of the study area.

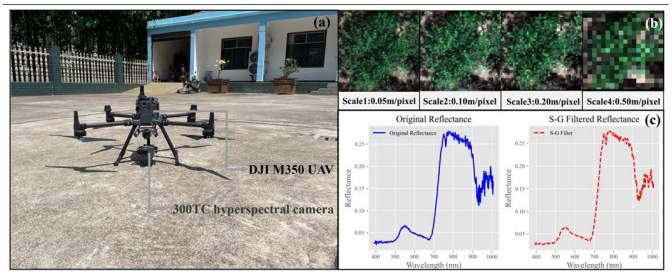


Fig. 3. (a) UAV hyperspectral system, (b) data in different spatial resolution, and (c) S-G filter process.

County is situated in the southern part of Hubei Province, at the confluence of the Yangtze River [28]. The region has a subtropical continental climate, with an annual sunshine duration ranging from 1803 to 1933 h. The average annual temperature is 16.6 °C. The area receives abundant rainfall, with an average annual relative humidity of 79%, providing favorable conditions for vegetation growth. As shown in Fig. 2, the sample plot is about 83 600 m^2. A total of 2069 *Populus* trees were planted in the study area, with each tree representing a unique genotype. All the *Populus* trees were planted in 2011, with an average diameter at breast height of 40.56 cm and average tree height of 35.68 m.

C. Data

1) UAV Hyperspectral Data Acquisition and Preprocessing: In this study, hyperspectral data were acquired between 11:00 and 12:00 local time on May 14, 2024 under cloud-free and windless conditions. The data were acquired using the 300TC hyperspectral camera mounted on a DJI M350 UAV. Table I provides a detailed description of the key sensor parameters and the flight experiment configuration. To mitigate bidirectional reflectance distribution function (BRDF) effects, UAV flights were conducted with a fixed flight direction, high forward and side overlap, and near-nadir

TABLE I
KEY PARAMETERS DESCRIPTION OF HYPERSPECTRAL AND LIDAR SENSORS

Sensors	Parameters
300TC hyperspectral camera	Wavelength range: 400 nm-1000 nm Spectral resolution: 2 nm Number of spectral channels: 300 Flight altitude: 100 m Spatial resolution of hyperspectral imagery: 0.045 m Forward overlap: 80% Side overlap: 70% Flight sortie: 2 Flight time: 11:00-12:00 local time on 14 May 2024
M350 DJ RTK Zenmuse L2 laser scanner system	Laser wavelength: 905 nm Laser pulse emission frequency: 240 kHz Flight altitude: 100 m Flight path: cross-shaped Flight speed: 5 m/s Side overlap: 70%

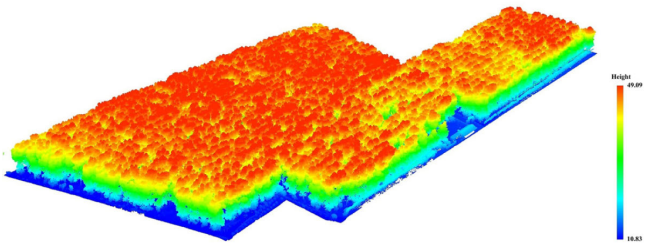


Fig. 4. Original LiDAR point clouds.

viewing geometry with view zenith angles of 4°. The raw hyperspectral data were processed using MegaCube software, including image mosaicking, radiometric correction, geometric correction, and orthorectification. The resulting hyperspectral reflectance imagery was then resampled to a spatial resolution of 0.05 m. Based on previous studies, spectral bands below 400 nm and above 900 nm were excluded due to increased noise and signal distortions [30]. Subsequently, the Savitzky–Golay (S–G) filter was applied to smooth the spectral curves with a window size of 5 (Fig. 3). After preprocessing, each pixel was represented by a smoothed reflectance spectrum consisting of 250 bands.

2) UAV LiDAR Data Acquisition and Preprocessing: As shown in Table I, the UAV LiDAR data and hyperspectral imagery were acquired on the same day using a M350 DJ RTK Zenmuse L2 laser scanner system. The raw data were processed to produce LiDAR point clouds with an average density of 1841 points/m^2(Fig. 4). The point clouds data was denoised and classified into ground and above-ground points using the LiDAR360 software. The DEM and DSM were generated using the Inverse Distance Weighting (IDW) interpolation method. The CHM data were produced based on the DEM and DSM data, with a spatial resolution resampled to 0.05 m.

3) Individual Tree Canopy Hyperspectral Dataset Production: First, the hyperspectral imagery is registered with the CHM data. This is achieved by geometrically correcting the hyperspectral images using orthophotography acquired from the same platform as the LiDAR data as a reference. Then,

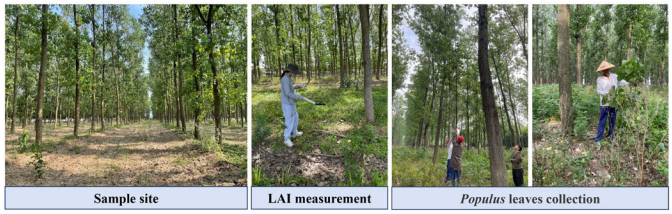


Fig. 5. LAI and leaves collections of individual *Populus* trees.

the hyperspectral imagery (Scale 1: 0.05 m) is resampled using cubic convolution interpolation to generate hyperspectral data at spatial resolutions of 0.10, 0.20, and 0.50 m [Fig. 3(b)]. The four spatial resolutions were determined considering the practical spatial resolution range of UAV-based hyperspectral systems and the typical crown size of *Populus* trees in the study area (approximately 6–8 m). Meanwhile, individual tree crown segmentation was conducted using the built-in algorithm implemented in LiDAR360. To quantitatively evaluate the segmentation accuracy, 50 individual tree crowns were randomly selected and compared with manually delineated reference crowns. The segmentation achieved an $F1$ score of 87.65%. Given that segmentation accuracy can substantially affect subsequent feature extraction, a visual inspection and manual refinement of the segmented crowns were further performed to ensure accurate delineation of crown boundaries. In this study, the spectral branch only uses the average value of all leaf pixels. The nonvegetation pixels were filtered using a normalized difference vegetation index (NDVI) threshold 0.7 [31], [32]. The spatial branch uses the hyperspectral imagery cubes as input. To reduce the influence of the background and boundary on feature extraction, a weighted mask is generated from the segmentation mask of the individual tree. The noncanopy areas are assigned a weight of 0, the boundary areas are assigned a weight of 0.5, and the canopy areas are assigned a weight of 1. This mask serves as the final layer of the hyperspectral cube, which is synchronized and downsampled with the spectral bands or VIs to ensure the accurate extraction of canopy features.

4) Field Survey and Laboratory Determination: This study collected LAI measurements from 480 individual *Populus* trees with different genotypes. As shown in Fig. 5, LAI values were measured using a LAI-2200C plant canopy analyzer (LI-COR Inc., Lincoln, NE, USA) on May 13 and 15, 2024. LAI measurement points were established at the north, south, east, and west directions of each tree. Three measurements were taken at each point, and the average value was used as the LAI for each tree. Following LAI data collection, leaves for LNC measurement were collected from 240 trees at four cardinal directions. The collected leaves were cleaned with distilled water, weighed, and stored at 4 °C for preservation. On each day after the data collection, the leaves were placed in an oven at 105 °C for 30 min, after which the temperature was reduced to 80 °C and the leaves were dried for 24 h until they reached a constant weight. The dried leaves were then ground into powder, and nitrogen content was determined using the Kjeldahl method.

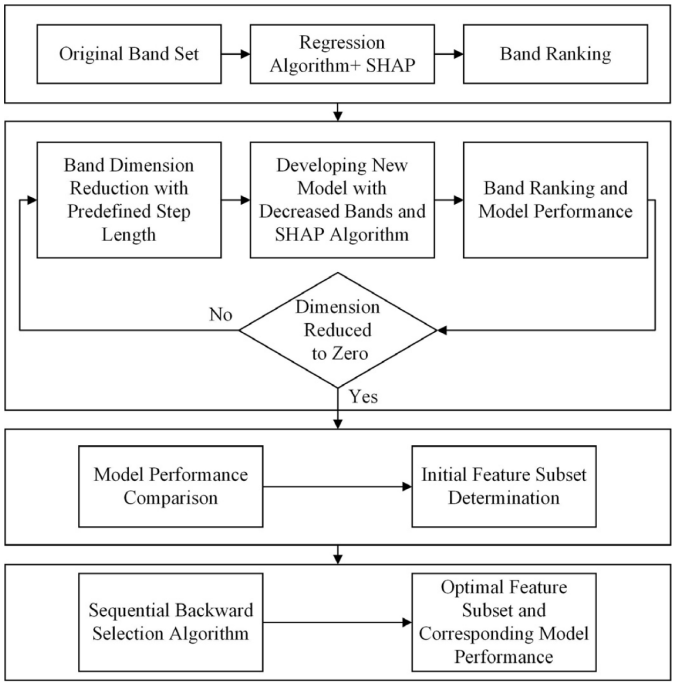


Fig. 6. Flowchart of hybrid feature selection algorithm.

D. Form of Input Features

This study compared three input features, including the original bands, feature-selected bands, and VIs. The feature selection algorithm and VIs calculation formulas are described in detail as follows.

1) Feature Selection Framework for Band Selection: This study used the hybrid feature selection framework proposed by Chen and Wang [33]. As shown in Fig. 6, this framework is combined with two steps. The first step is to determine the initial feature subset by a stepwise dimensionality reduction process, and the second step is to identify the optimal feature subset using the wrapper method. In this framework, the Shapley Additive exPlanations (SHAP) algorithm was introduced to calculate the contribution of each band, and the reranking strategy was used to handle the dynamic change of the feature subset. As illustrated in Fig. 6, a regression algorithm was required within this framework. Given that the feature selection in this study focused on eliminating redundant features rather than extracting deep features, the traditional machine learning algorithm support vector regression (SVR) was employed. This algorithm is well-suited for handling shallow features and modeling nonlinear relationships by mapping input data into a higher-dimensional space using kernel functions. The radial basis function (RBF) kernel was used in the SVR algorithm. Three key hyperparameters, including regularization parameters (C), epsilon (ε), and gamma (γ), were optimized through grid search. Sequential backward selection (SBS) was used as a wrapper to identify the optimal feature subset. The feature selection process began with all available features, and the SBS method iteratively eliminated the least informative ones. Model performance was reevaluated after each removal using root mean square error (RMSE) as

TABLE II
NARROW-BAND VIS USED FOR DEVELOPING LAI AND LNC ESTIMATION MODELS

VIs	Formula	Application	Ref
Carte1	R_{695}/R_{421}	LNC	[34]
Carte2	R_{695}/R_{761}	LNC	[34]
Carte3	R_{605}/R_{761}	LNC	[34]
Carte4	R_{711}/R_{761}	LNC	[34]
Carte5	R_{695}/R_{671}	LNC	[34]
Carotenoid/Chlorophyll Ratio Index (CCRI)	$[(R_{721}-R_{521})\times R_{705}]/[(R_{751}-R_{705})\times R_{521}]$	LNC	[35]
Red Edge Chlorophyll Index (CI_{re})	(R_{801}/R_{721})-1	LAI and LNC	[36]
Datt1	$(R_{851}-R_{711})/(R_{851}-R_{681})$	LNC	[37]
Datt2	R_{851}/R_{711}	LNC	[37]
Datt3	R_{755}/R_{705}	LNC	[37]
Double-peak Canopy Nitrogen Index (DCNI)	$[(R_{721}-R_{701})\times(R_{721}-R_{671}+0.03)]/(R_{701}-R_{671})$	LNC	[38]
Enhanced Vegetation Index (EVI)	$2.5\times[(R_{801}-R_{671})/(R_{801}-6\times R_{671}-7.5\times R_{475}+1)]$	LAI	[39]
Enhanced Vegetation Index2 (EVI2)	$2.5\times[(R_{801}-R_{671})/(R_{801}+2.4\times R_{671}+1)]$	LAI	[40]
GI (Greenness Index)	R_{554}/R_{677}	LNC	[41]
Green Normalized Difference Vegetation Index (GNDVI)	$(R_{751}-R_{551})/(R_{751}+R_{551})$	LNC	[42]
Modified Chlorophyll Absorption Ratio Index (MCARI)	$[(R_{701}-R_{671})-0.2\times(R_{701}-R_{551})]\times(R_{701}/R_{671})$	LAI	[43]
Modified Chlorophyll Absorption Ratio Index1 (MCARI1)	$1.2\times[2.5\times(R_{801}-R_{671})-1.3\times(R_{801}-R_{551})]$	LAI	[44]
Modified Chlorophyll Absorption Ratio Index2 (MCARI2)	$1.5\times[2.5\times(R_{801}-R_{671})-1.3\times(R_{801}-R_{551})]\times[(2\times R_{801}+1)^2-(6\times R_{801}-5\times R_{671}{}^{0.5})]^{0.5}$	LAI	[44]
Modified Normalized Difference Vegetation Index$_{705}$ (mNDVI$_{705}$)	$(R_{751}-R_{705})/(R_{751}+R_{705}-2\times R_{445})$	LAI	[45]
Modified Soil-Adjusted Vegetation Index (MSAVI)	$0.5\times\{2\times R_{801}+1-[(2\times R_{801}+1)^2-8\times(R_{801}-R_{671})]^{0.5}\}$	LAI	[46]
Modified Simple Ratio Index (MSR)	$(R_{801}/R_{671}-1)/(R_{801}/R_{671}+1)^{1/2}$	LAI	[47]
Modified Simple Ratio$_{705}$ (mSR$_{705}$)	$(R_{705}-R_{445})/(R_{705}+R_{445})$	LAI	[45]
MERIS terrestrial chlorophyll index (MTCI)	$(R_{754}-R_{709})/(R_{709}-R_{681})$	LAI and LNC	[48]
Modified Triangular Vegetation Index1 (MTVI1)	$1.2\times[1.2\times(R_{801}-R_{551})-2.5\times(R_{671}-R_{551})]$	LAI and LNC	[44]
Modified Triangular Vegetation Index2 (MTVI2)	$1.5\times[1.2\times(R_{801}-R_{551})-2.5\times(R_{671}-R_{551})]\times(2\times R_{801}+1)^2-(6\times R_{801}-5\times R_{671})-0.5$	LAI and LNC	[44]
Normalized Difference Red Edge Index (NDRE)	$(R_{781}-R_{721})/(R_{781}+R_{721})$	LAI and LNC	[49]
Normalized Difference Vegetation Index (NDVI)	$(R_{801}-R_{681})/(R_{801}+R_{681})$	LAI and LNC	[50]
Normalized Difference Vegetation Index$_{705}$ (NDVI$_{705}$)	$(R_{751}-R_{705})/(R_{751}+R_{705})$	LAI and LNC	[49]
Normalized Pigment Chlorophyll Ratio Index (NPCI)	$(R_{671}-R_{461})/(R_{671}+R_{461})$	LNC	[51]
Normalized Phaeophytinization Index (NPQI)	$(R_{415}-R_{435})/(R_{415}+R_{435})$	LNC	[52]
Optimized Soil-Adjusted Vegetation Index (OSAVI)	$1.16\times(R_{801}-R_{671})/(R_{801}+R_{671}+0.16)$	LAI	[53]
Pigment-Specific Normalized Difference (PSND)	$(R_{801}-R_{471})/(R_{801}+R_{471})$	LNC	[54]
Plant Senescence Reflectance Index (PSRI)	$(R_{681}-R_{501})/R_{751}$	LAI and LNC	[55]
Photosynthetic Vigour Ratio (PVR)	$(R_{551}-R_{651})/(R_{551}+R_{651})$	LNC	[56]
Red-Edge Vegetation Stress Index (RVSI)	$[(R_{713}+R_{753})/2]-R_{733}$	LAI and LNC	[57]
Soil-Adjusted Vegetation Index (SAVI)	$1.5\times(R_{801}-R_{671})/(R_{801}+R_{671}+0.5)$	LAI	[58]
Structure Insensitive Pigment Index (SIPI)	$(R_{801}-R_{445})/(R_{801}-R_{681})$	LAI	[59]
Simple Ratio$_{[700,670]}$ (SR$_{[700,670]}$)	R_{801}/R_{681}	LAI and LNC	[60]
Simple Ratio$_{[750,550]}$ (SR$_{[750,550]}$)	R_{701}/R_{671}	LAI and LNC	[60]
Simple Ratio$_{[750,700]}$ (SR$_{[750,700]}$)	R_{751}/R_{701}	LAI and LNC	[60]
Simple Ratio$_{[750,710]}$ (SR$_{[750,710]}$)	R_{753}/R_{691}	LAI and LNC	[60]
Simple Ratio$_{[752,690]}$ (SR$_{[752,690]}$)	R_{751}/R_{551}	LAI and LNC	[60]
Simple Ratio$_{[800,680]}$ (SR$_{[800,680]}$)	R_{751}/R_{711}	LAI and LNC	[60]
Simple Ratio Pigment Index (SRPI)	R_{431}/R_{681}	LNC	[35]
Transformed Chlorophyll Absorption in the Reflectance Index (TCARI)	$3\times[(R_{701}-R_{671})-0.2\times(R_{701}-R_{551})(R_{701}/R_{671})]$	LAI and LNC	[61]
Transformed Chlorophyll Absorption in the Reflectance Index2 (TCARI2)	$3\times[(R_{751}-R_{705})-0.2\times(R_{751}-R_{551})(R_{751}/R_{705})]$	LAI and LNC	[62]
Triangular vegetation index (TVI)	$0.5\times[120\times(R_{751}-R_{551})-200\times(R_{671}-R_{551})]$	LAI and LNC	[63]
Optimal Vegetation Index (VI$_{opt}$)	$1.45\times((R_{801})^2+1)/(R_{671}+0.45)$	LNC	[64]
Photochemical Reflectance Index (PRI)	$(R_{531}-R_{571})/(R_{531}+R_{571})$	LNC	[65]

TABLE II
(Continued.) NARROW-BAND VIS USED FOR DEVELOPING LAI AND LNC ESTIMATION MODELS

VOG1	R_{741}/R_{721}	LNC	[66]
VOG2	$(R_{735}-R_{747})/(R_{715}+R_{727})$	LNC	[66]
VOG3	$(R_{735}-R_{747})/(R_{715}+R_{721})$	LNC	[66]

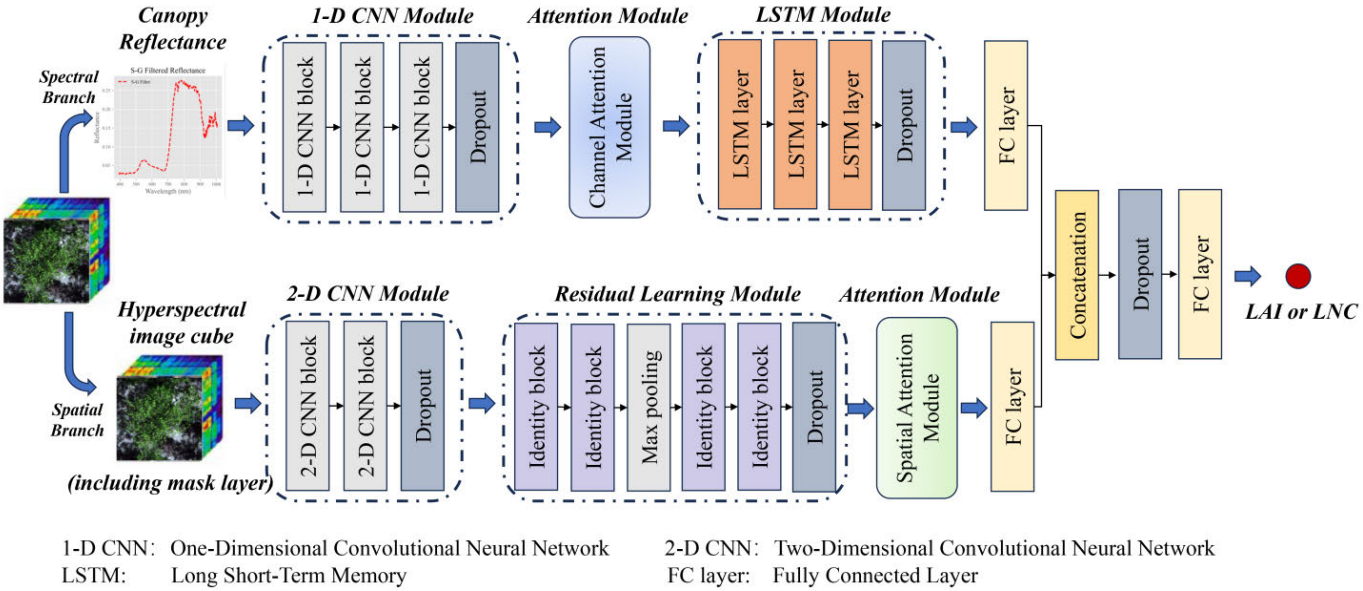


Fig. 7. Structure of dual-branch DL network.

the evaluation metric. This process continued until no features remained in the subset.

2) Sensitive VIs for LAI and LNC Estimation: VIs can reduce interference factors and refine target information compared to the full spectrum. Narrowband VIs generated by hyperspectral data offer greater sensitivity to subtle variations in vegetation biophysical and biochemical properties than broadband VIs. Therefore, narrowband VIs are widely used for estimating LAI and LNC. Table II presents several VIs used in previous studies. In this study, 40 VIs were used for LNC estimation, while 30 VIs were used for LAI estimation. Given that the hyperspectral bands acquired in this study are odd-numbered at 2 nm intervals, the even-numbered wavelengths in the original formula were substituted with the nearest available odd-numbered counterparts (e.g., 600 nm was replaced with 601 nm).

E. Dual-Branch DL Network

The dual-branch architecture is a synergistic DL framework that enables differentiated yet complementary feature learning by assigning distinct subnetworks to process spectral and spatial information, respectively. To overcome the limitations of single-branch models that often suffer from early-stage feature entanglement, this study designed a multimodel dual-branch architecture which process the spectral and spatial feature of hyperspectral data, respectively. As shown in Fig. 7, the inputs to the spectral and spatial branches are the average canopy reflectance and the hyperspectral image cube of individual *Populus* trees, respectively. The spectral branch comprises a one-dimension CNN (1-D CNN) module, a channel attention module, and an LSTM module. Similarly, the spatial branch includes a 2-D CNN, residual learning, and spatial attention modules. In the spectral branch, the 1-D CNN module serves as a feature extractor, transforming the original reflectance data into a more informative representation emphasizing key spectral patterns. This is followed by a channel attention module, which dynamically recalibrates the importance of spectral bands to enhance discriminative spectral responses. Subsequently, the extracted features are fed into the LSTM module to capture long-range spectral dependencies and sequential patterns in the canopy reflectance. In parallel, the spatial branch begins with a 2-D CNN module for initial feature extraction. The resulting shallow features are then passed into the residual learning module, consisting of four identity blocks and max pooling and dropout layers. These identity blocks mitigate gradient vanishing issues via skip connections, facilitating deeper network training. The down-sampling process via max pooling layer enables hierarchical feature extraction, and the dropout layer is used to enhance generalization, which is particularly crucial for hyperspectral applications with limited training samples and high-dimensional data. Following residual learning, the spatial attention module adaptively emphasizes informative regions by generating attention maps and refining feature representations based on spatial context. The outputs from both branches are then concatenated via a concatenation layer. A subsequent dropout layer is employed to prevent overfitting and ensure

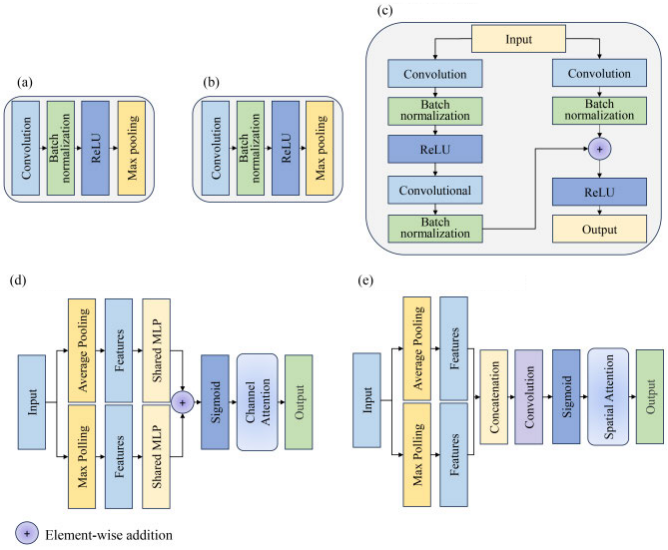


Fig. 8. Structure of 1-D CNN block, 2-D CNN block, identity block, channel attention module, and spatial attention module.

robust model regularization. Finally, a fully connected layer captures complex interactions between the spectral and spatial features extracted from the dual branches.

Fig. 8 illustrates the structures of the 1-D CNN block, 2-D CNN block, and identity block. In each CNN block, every convolutional layer is followed by a batch normalization (BN) layer and a rectified linear unit (ReLU) activation function to ensure stable and effective feature extraction. A max pooling layer is subsequently applied to reduce the spatial dimensions and computational complexity. The identity block serves as a key component of the residual learning module. It consists of two convolutional layers, each followed by a BN layer to promote stable training. A convolutional shortcut connection directly links the input and output, enabling the model to learn residual mappings and facilitating the flow of gradients during backpropagation. This study incorporates a channel attention mechanism (CAM) and a spatial attention mechanism (SAM) into the spectral and spatial branches, respectively. The CAM is designed to emphasize informative spectral channels by computing attention scores for each channel and adjusting the feature map accordingly, assigning higher weights to more important channels. In contrast, the SAM enables the model to adaptively learn attention weights for different spatial regions, allowing it to focus on critical areas of the image while suppressing less relevant regions. The structural designs of both CAM and SAM are based on the convolutional block attention module (CBAM) proposed in a previous study [66]. As shown in Fig. 8, the CAM compresses input features into channel descriptors using average and max pooling, passing through a shared multilayer perceptron (MLP) to generate activation weights. These weights are applied via channel-wise multiplication to enhance informative features and suppress redundant ones. The SAM, on the other hand, compresses channel information by computing average-pooled and max-pooled feature maps across channels. These maps are concatenated and passed through a convolutional layer with a sigmoid activation function to produce a spatial attention map. This map is then element-wise multiplied with the input features to highlight important spatial locations and diminish irrelevant background information.

In this study, the input data for each branch were carefully selected, considering both spatial resolution (Scale 1: 0.05 m; scale 2: 0.10 m; scale 3: 0.20 m; scale 4: 0.30 m) and input feature form (all spectral bands, feature-selected bands, and VIs). Accordingly, twelve spatial resolution and input form combinations were tested and compared within each branch. Based on the evaluation results, the optimal combination for each branch was determined and subsequently used to construct the dual-branch models for estimating the LAI and LNC of individual *Populus* trees.

F. Comparative Experiments

To validate the effectiveness of the proposed dual-branch LAI and LNC estimation models, two widely used DL architectures, namely 3D-CNN and CNN–LSTM, were selected as comparative models. The 3D-CNN represents a classical spectral–spatial joint modeling paradigm, in which spectral and spatial features are implicitly coupled through 3-D convolution, whereas the CNN–LSTM explicitly models spectral sequential dependencies by integrating convolutional feature extraction with recurrent learning. Both models aim to exploit spectral–spatial information, which is consistent with the fundamental modeling objective of the proposed dual-branch framework. The details of each experiment follow.

1) 3-D CNN: The 3-D CNN model is an important DL structure for processing 3-D data, such as hyperspectral imagery and time series remote sensing data. In a 3-D CNN model, the input data is typically represented as a 3-D volume, with dimensions (w, h, d), where m represents the width, n represents the height, and d represents the depth. When applying the 3-D CNN model for estimation, a patch defined by a window size of $m \times m \times l$ is used. As shown in Fig. 9, this study developed 3-D CNN estimation models using four convolutional blocks. Each block contains two convolutional layers, one BN layer, one max pooling layer, and one ReLU active function for effective and stable feature extraction. The dropout layer was incorporated into this network to prevent overfitting and improve model generalization.

2) CNN-LSTM: The CNN-LSTM model is another hybrid DL architecture for processing data in the spatial and temporal domains. In the CNN-LSTM architecture, the CNN blocks are used to extract the hierarchical spatial features from UAV hyperspectral data to capture local spatial patterns. Then the LSTM layers process the sequential dependencies of these extracted features and develop the sequential and multivariate estimation model. The integration of CNN and LSTM allows this architecture to better capture the spatial and temporal or sequential dynamics inherent in the hyperspectral data [67]. As shown in Fig. 10, the convolutional block contains one convolutional layer, one BN layer, and one max pooling layer. Then, the feature vector was flattened and used as the input for the LSTM layers, which was further computed through the fully connected layer to estimate the final LAI or LNC result. To prevent overfitting and increase the model's efficiency, this

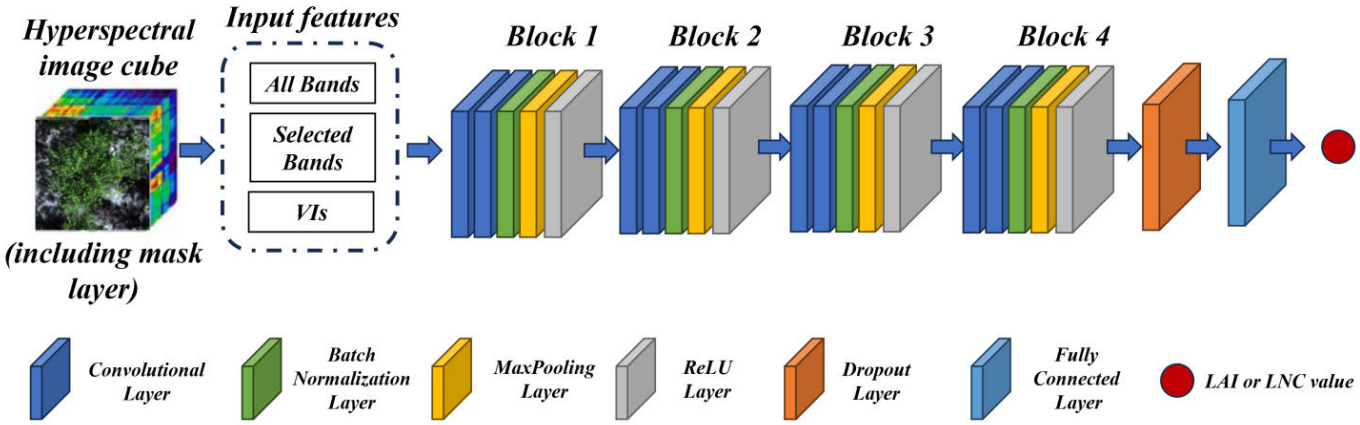


Fig. 9. Structure of 3-D CNN model.

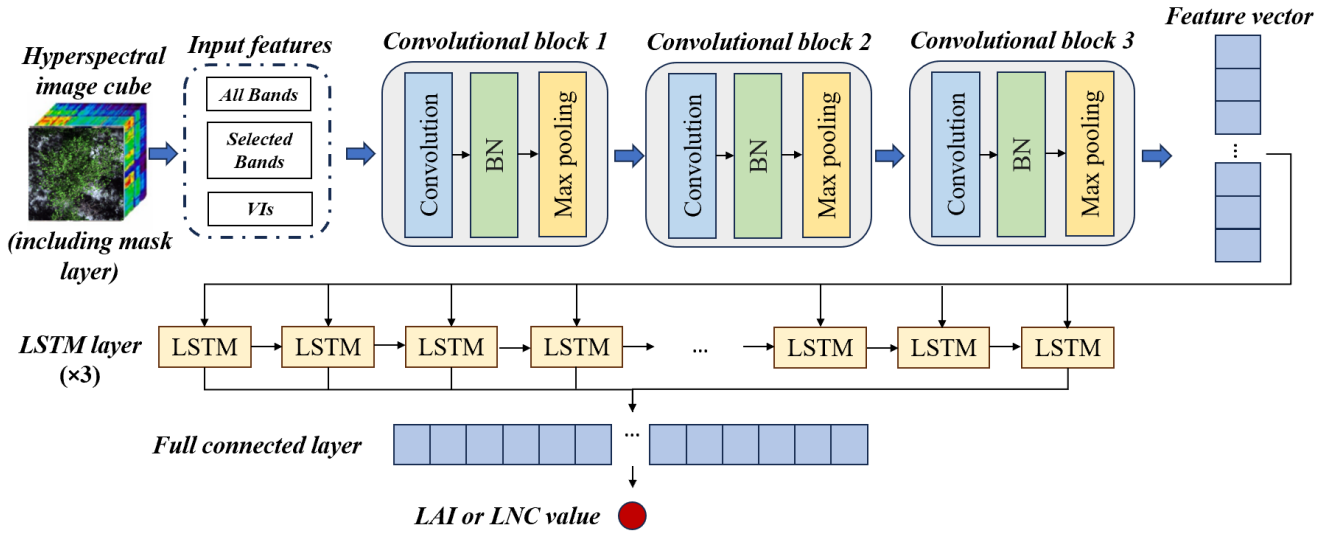


Fig. 10. Structure of CNN-LSTM model.

study used dropout in the fully connected layer with a value of 0.5.

Based on the three hybrid DL architectures, a comprehensive evaluation was conducted in terms of estimation accuracy and computational efficiency. After each model determined its optimal scale and feature form, robustness to noise was further assessed using three hybrid models and two single branch models, including spectral and spatial branches. To simulate realistic background noise, the Gaussian mixture model (GMM) was applied. Specifically, 500 samples were selected for each background component, including branches, shadows, and soil, to fit a GMM, from which synthetic noise samples were generated. These samples were then used to randomly replace 3%, 5%, and 10% of the total canopy pixels. This procedure allowed the robustness of the models to be systematically evaluated under increasing noise intensities while preserving the statistical characteristics of the original background components. All models were trained and tested under the same hardware conditions (Intel Core i9 @ 2.00 GHz, NVIDIA RTX A4000, 128 GB RAM) to ensure a fair comparison of computational performance.

G. Data Augmentation and Model Evaluation

In this study, 2/3 of the LAI and LNC samples were randomly selected for developing the training dataset, and 1/3 of the samples were used as the testing dataset. To mitigate the risk of overfitting, this study applied data augmentation to the training dataset. Various data augmentation techniques were used for estimation tasks, such as affine transformations (e.g., rotation, resizing, etc.), elastic deformations (e.g., shape variations), and pixel-level modifications (e.g., adding Gaussian noise) [68], [69]. Since the DL architecture proposed in this study contains a spectral branch (reflectance curve) and a spatial branch (hyperspectral image cube), two data augmentation techniques were applied, including affine transformation and pixel-level modification. First, the hyperspectral image cubes were rotated at four angles: 0°, 90°, 180°, and 270°. Then, 1% Gaussian white noise was added to each band in each pixel. Therefore, the samples of LAI and LNC training datasets increased to 1600 and 800, respectively. The samples of testing datasets of LAI and LNC contain 160 and 80, respectively. To reduce the potential randomness introduced by a single

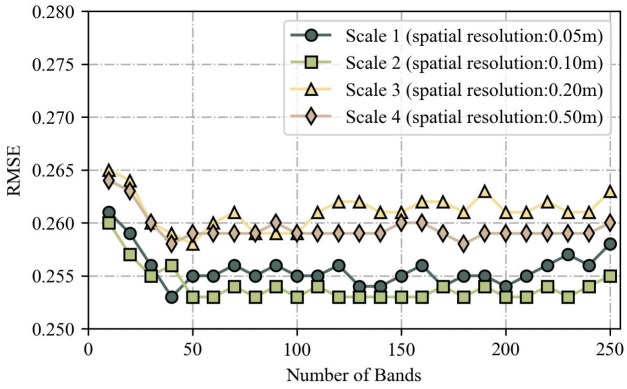


Fig. 11. Mean RMSE values of LAI estimation models across dimensions and scales using the SVR algorithm.

data partition, the above random train–test split was repeated 10 times, and the model was independently trained and evaluated for each split. The final performance metrics were reported as the mean and standard deviation across the 10 runs. The model performance evaluation indicators used in this study include the coefficients of determination (R^2), and RMSE. The calculation formulas for all the relevant indicators are provided below

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - y_i'\right)^2}{\sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \left(y_i' - y_i\right)^2}$$

where y_i, $\bar{y}$, and y_i' represent the measured LAI (or LNC), mean value of measured LAI (or LNC) and the estimated LAI (or LNC).

III. Results

A. Band Selection Results in Different Scales

1) Band Selection Results of LAI Estimation Models: This study used a hybrid feature selection algorithm to identify the optimal band combinations of LAI estimation models in each scale. Fig. 11 shows the change in the mean RMSE value of LAI estimation models when the number of bands is reduced according to their SHAP contribution. The LAI estimation models at Scales 1, 2, 3, and 4 achieved their highest estimation accuracy at 40, 50, 50, and 40 dimensions, respectively, with RMSE values of 0.253, 0.253, 0.258, and 0.258. These bands were defined as the initial subsets for each scale. In Scales 1 and 2, the changes in RMSE are minimal during the process of reducing the bands from the full dimension to the initial subsets. In contrast, the RMSE curves of Scales 3 and 4 show a decrease when the band number is reduced from 130 to 50. Following the hybrid feature selection principle, the SBS algorithm was applied to each initial band subset to search for the optimal band subset.

Fig. 12 shows the performance (including the mean value and standard deviation of each evaluation indicators) of LAI estimation models using the optimal band subsets and SVR algorithm at each scale. Among four models, the model of Scale 2 achieved the highest estimation accuracy, with R^2 of 0.754 ± 0.006, RMSE of 0.236 ± 0.003. In contrast, the model of Scale 4 achieved the lowest accuracy of 0.730 ± 0.011, RMSE of 0.250 ± 0.008. The results in Fig. 12 show that the accuracy difference between Scale 1 and Scale 2 is minimal, with an RMSE difference of 0.007. Similarly, the accuracy difference between Scale 3 and 4 is also slight, with an RMSE difference of 0.003.

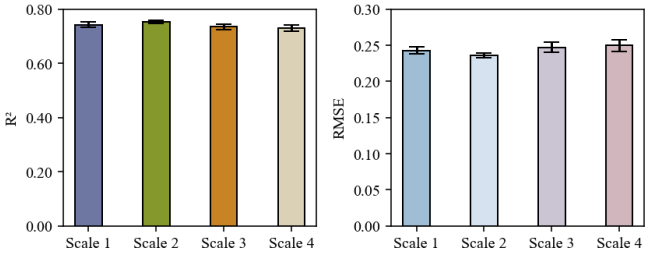


Fig. 12. Optimal LAI estimation model performance of different scales using SVR and hybrid feature selection algorithms.

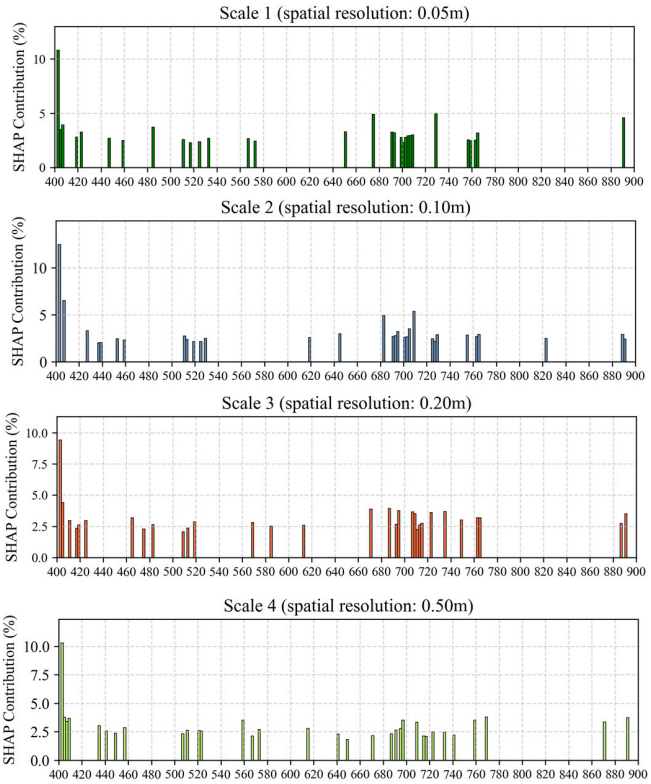


Fig. 13. SHAP contributions of selected bands in LAI estimation model in each scale.

Fig. 13 illustrates the band selection results and their SHAP contributions in each model. The optimal band subsets for Scales 1, 2, 3, and 4 consist of 30, 31, 31, and 34 bands, respectively. These bands exhibit a similar distribution across the 400–900 nm range. For example, the selected bands in each scale were predominantly concentrated in the red-edge (RE) bands, ranging from 680 to 750 nm. In contrast, the red bands were less frequently selected due to the saturation effect.

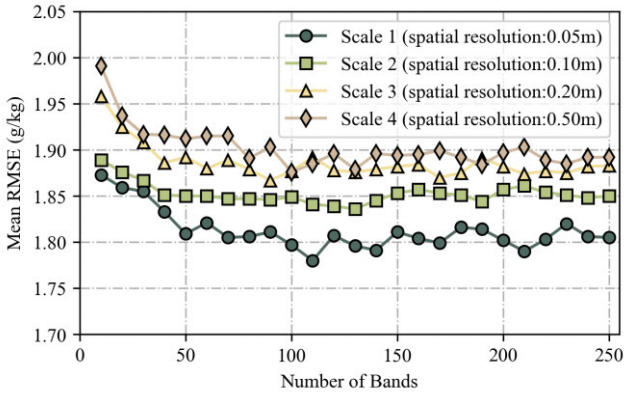


Fig. 14. Mean RMSE values of LNC estimation models across dimensions and scales using the SVR algorithm.

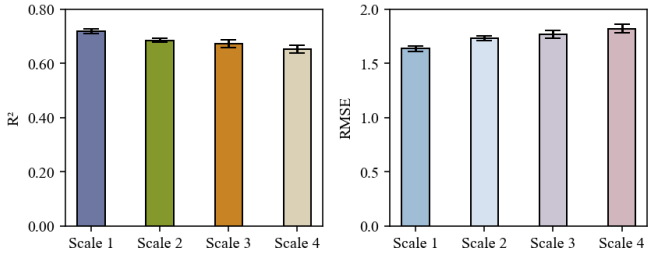


Fig. 15. Optimal LNC estimation model performance of different scales using SVR algorithm.

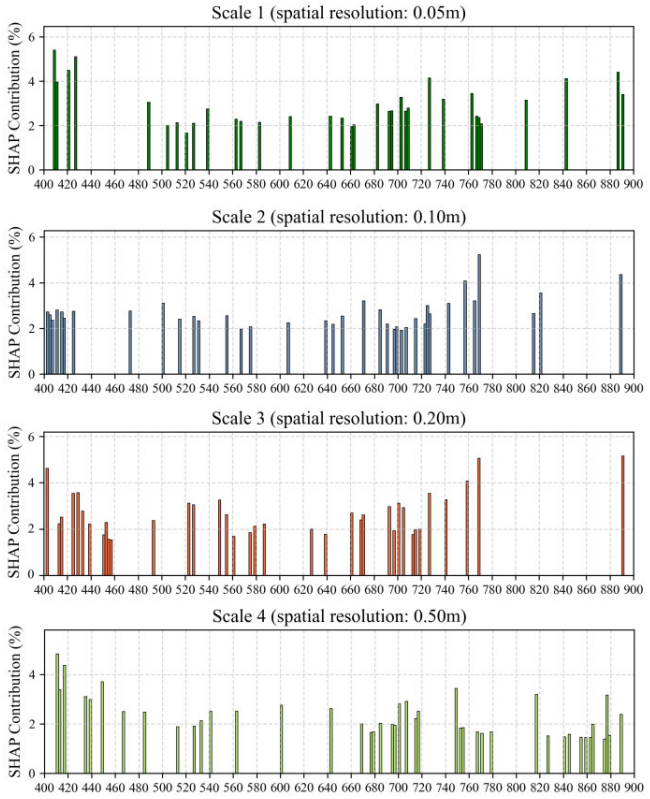


Fig. 16. SHAP contributions of selected bands in LNC estimation model in each scale.

The blue band with a central wavelength of 403 nm in each scale achieved the highest SHAP contribution, with 10.71%, 12.45%, 9.43%, and 10.32%, respectively.

2) Band Selection Results of LNC Estimation Models: Fig. 14 shows the initial band selection results for four LNC estimation models. Compared to the LAI estimation results, the changes in mean RMSE values are more pronounced as the number of bands decreases. The RMSE values of Scale 1 in each dimension are lower than the results of other scales, which indicates that higher spatial resolution can improve the LNC estimation accuracy. The LNC estimation models at Scales 1, 2, 3, and 4 achieved their highest estimation accuracy at 110, 130, 90, and 100 dimensions, with RMSE values of 1.780 $\pm$ 0.031 g/kg, 1.836 $\pm$ 0.044 g/kg, 1.867 $\pm$ 0.062 g/kg, and 1.870 $\pm$ 0.060 g/kg. According to Fig. 14, the difference between the results obtained from Scale 3 and Scale 4 is minimal, and the variation in the RMSE curves follows a similar pattern. This result indicated that when the spatial resolution is below 0.20 m, the influence of noise (e.g., shadows, soil) on reflectance becomes more pronounced.

Fig. 15 shows the LNC estimation performance of each optimal model corresponding to each scale. The Scale 1 model achieved the highest LNC estimation accuracy with an R^2 of 0.714 $\pm$ 0.009 and RMSE of 1.636 $\pm$ 0.025 g/kg. The model of Scale 4 achieved the lowest accuracy with an R^2 of 0.653 $\pm$ 0.016 and an RMSE of 1.821 $\pm$ 0.041 g/kg. The RMSE difference between Scale 1 and Scale 2 models is 0.097 g/kg. Similarly, the accuracy difference between Scale 3 and 4 is also slight, with an RMSE difference of 0.053 g/kg.

Fig. 16 presents the band selection results and their corresponding SHAP contributions for each model. The optimal band subsets for Scales 1, 2, 3, and 4 consist of 34, 37, 37, and 44 bands, respectively. The results indicate that the sensitive bands and their SHAP contributions change accordingly as the spatial scale varies. For Scales 1, 2, and 3, the selected bands are predominantly concentrated in the Red and RE (620–750 nm) spectral range, with fewer than five NIR bands selected. However, in Scale 4, twelve NIR bands are selected. The optimal band subset for Scale 1 includes four Blue bands, each exhibiting a SHAP contribution greater than 4%. In comparison, the optimal band subset for Scale 3 includes eight Blue bands, each showing a SHAP contribution lower than 3%. Compared to the results of the LAI estimation models, the accuracy and band selection results of LNC estimation models are more significantly affected by the spatial resolution of hyperspectral data and require more bands to achieve accurate estimations.

B. LAI and LNC Estimation Result of Dual-Branch DL Networks

1) Selection Results of Spectral Branch: This study used all bands, selected bands, and VIs in each scale as inputs for developing the spectral branch of the LAI and LNC estimation model. Since two evaluation indicators show the same trends, this study only used R^2 to illustrate the comparison results. As shown in Table III, regardless of the input feature forms, the LAI estimation models derived from Scale 2 consistently

TABLE III

MEAN AND STANDARD DEVIATION OF R^2 VALUES OF LAI AND LNC ESTIMATION MODELS USED FOR DEVELOPING SPECTRAL BRANCH

Model	Inputs	Scale 1	Scale 2	Scale 3	Scale 4
LAI estimation model	All bands	0.804 (± 0.006)	0.814 (± 0.006)	0.801 (± 0.008)	0.784 (± 0.008)
	Selected bands	0.813 (± 0.006)	0.820 (± 0.006)	0.800 (± 0.006)	0.779 (± 0.009)
	VIs	0.784 (± 0.008)	0.790 (± 0.007)	0.782 (± 0.007)	0.765 (± 0.009)
LNC estimation model	All bands	0.790 (± 0.005)	0.762 (±0.007)	0.741 (±0.009)	0.725 (±0.010)
	Selected bands	0.795 (± 0.006)	0.765 (± 0.008)	0.733 (± 0.007)	0.716 (± 0.011)
	VIs	0.787 (± 0.008)	0.754 (± 0.008)	0.724 (± 0.009)	0.712 (± 0.010)

TABLE IV

MEAN AND STANDARD DEVIATION OF R^2 VALUES OF LAI AND LNC ESTIMATION MODELS USED FOR DEVELOPING SPATIAL BRANCH

Model	Inputs	Scale 1	Scale 2	Scale 3	Scale 4
LAI estimation model	All bands	0.785 (± 0.008)	0.788 (± 0.008)	0.775 (± 0.008)	0.759 (± 0.007)
	Selected bands	0.792 (± 0.008)	0.799 (± 0.010)	0.776 (± 0.007)	0.760 (± 0.009)
	VIs	0.805 (± 0.006)	0.817 (± 0.007)	0.788 (± 0.007)	0.772 (± 0.008)
LNC estimation model	All bands	0.777 (± 0.009)	0.726 (± 0.009)	0.721 (± 0.008)	0.713 (± 0.005)
	Selected bands	0.781 (± 0.007)	0.730 (± 0.009)	0.704 (± 0.011)	0.699 (± 0.006)
	VIs	0.787 (± 0.007)	0.736 (± 0.009)	0.707 (± 0.010)	0.700 (± 0.006)

outperform those from other scales. This result is consistent with the result shown in Fig. 12. Therefore, these results indicate that Scale 2 (0.10 m/pixel) is the optimal spatial resolution for constructing the spectral branch of the LAI estimation model in this study. As the spatial resolution increases, the estimation accuracy obtained using selected bands gradually surpasses that of the all-bands-based models. Meanwhile, the model accuracy based on VIs remains the lowest across all scales. According to the model comparison, selected bands in Scale 2 (with R^2 of 0.820 $\pm$ 0.006 and RMSE of 0.203 $\pm$ 0.011) were determined as the input for developing the spectral branch of the LAI estimation model. Table III also shows the LNC estimation results when using different input feature forms in each scale. The results indicated that Scale 1 (0.05 m/pixel) is the optimal spatial resolution for developing an LNC estimation model in this study. When the spatial resolution is below 0.10 m, the highest LNC estimation accuracy is achieved using all bands. However, when the spatial resolution is higher than 0.10 m, the accuracy obtained using selected bands surpasses all bands. In contrast, the accuracy obtained using VIs is the lowest across all scales. Therefore, the selected bands of Scale 1 (with R^2 of 0.795 $\pm$ 0.006, RMSE of 1.400 $\pm$ 0.015 g/kg) were determined as the input for developing the spectral branch of the LNC estimation model for an individual *Populus* tree.

2) Selection Results of Spatial Branch: Table IV shows the performances of the LAI and LNC models developed for the spatial branch. The models constructed using the three input forms achieved the highest accuracy when Scale 2 data was used. This result indicated that Scale 2 (0.10 m/pixel) is the optimal spatial resolution for developing the spatial branch of the LAI estimation model. This conclusion is consistent with the findings obtained from the spectral branch. However, in contrast to the spectral branch results, the model accuracy in the spatial branch was higher for VIs than for all bands and selected bands when using Scale 1, Scale 2, and Scale 3 data. Based on the model comparison results, VIs in Scale 2 (with R^2 of 0.817 $\pm$ 0.007, RMSE of 0.205 $\pm$ 0.009) were determined as the optimal input for developing the spectral branch of the LAI estimation model. Table IV also shows the performance of the LNC estimation model developed using different input forms in each scale. Similar to the results in the spectral branch, the models developed by Scale 1 outperform models in other scales. These results all indicate that high spatial resolution significantly improves the accuracy of LNC estimation for individual *Populus* trees. In Scales 3 and 4, the all-bands-based models achieved the highest R^2 values, and the selected bands achieved the lowest R^2 values. While in Scales 1 and 2, the VIs-based models achieved the best model performances, the all-bands-based models show the lowest R^2 values. This result suggests that, at lower spatial resolutions, additional spectral bands can enhance LNC estimation. In contrast, high-precision LNC estimation can be achieved at higher spatial resolutions by utilizing a subset of sensitive bands or VIs derived from them. Finally, the VIs of Scale 1 (with R^2 of 0.787 $\pm$ 0.007 and RMSE of 1.426 $\pm$ 0.017 g/kg) were determined as the input for developing the spatial branch of the LNC estimation model for an individual *Populus* tree.

3) Estimation Results of Dual-Branch DL Architecture: Based on the scale and input selections in each branch, the

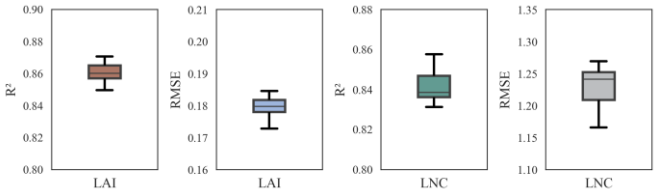


Fig. 17. Results of ten modeling runs for the LAI and LNC estimation models using dual-branch DL architecture.

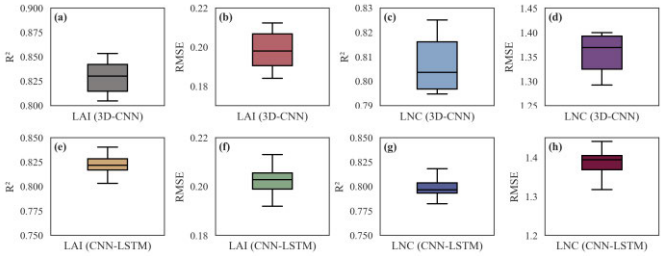


Fig. 18. Results of ten modeling runs for the LAI and LNC estimation models using 3-D CNN and CNN-LSTM architectures. (a) R^2 of ten LAI modeling runs based on 3-D CNN and selected bands of Scale 2. (b) RMSE of ten LAI modeling runs based on 3-D CNN and selected bands of Scale 2. (c) R^2 of ten LNC modeling runs based on 3-D CNN and selected bands of Scale 1. (d) RMSE of ten LN modeling runs based on 3-D CNN and selected bands of Scale 1. (e) R^2 of ten LAI modeling runs based on CNN-LSTM and selected bands of Scale 2. (f) RMSE of ten LAI modeling runs based on CNN-LSTM and selected bands of Scale 2. (g) R^2 of ten LNC modeling runs based on CNN-LSTM and selected bands of Scale 1. (h) RMSE of ten LNC modeling runs based on CNNLSTM and selected bands of Scale 1.

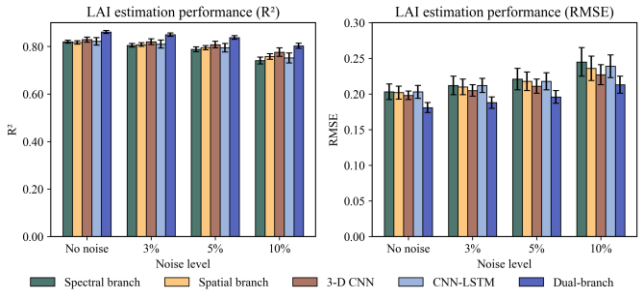


Fig. 19. LAI estimation accuracy of three hybrid architectures and two branch-based models under different noise levels.

dual-branch LAI and LNC estimation models were developed. As shown in Fig. 17, the LAI estimation model achieved a performance with R^2 of 0.861 $\pm$ 0.006 and RMSE of 0.181 $\pm$ 0.007. Compared with the results of the spectral and spatial branches, the RMSE of the dual-branch LAI estimation model has been reduced by 12.15% and 13.26%. The LNC estimation model achieved performance with R^2 of 0.842 $\pm$ 0.009 and RMSE of 1.230 $\pm$ 0.034 g/kg. This dual-branch LNC estimation model has reduced RMSEs of 13.82% and 15.94% compared with the spectral and spatial branches, respectively. Overall, the above results indicate that the dual-branch models consistently outperform the corresponding single-branch models for both LAI and LNC estimation, with a notable improvement in accuracy while maintaining comparable variability across repeated experiments. Importantly, the standard deviations of R^2 and RMSE show no substantial increase relative to the single-branch counterparts, suggesting that the performance gains are not accompanied by a loss of model stability. This implies that the observed improvements cannot be simply attributed to an increase in overall model capacity. Instead, they stem from the complementary and synergistic fusion of spectral and spatial information, which enables the dual-branch architecture to capture more discriminative and representative features. Consequently, the dual-branch design effectively enhances the feature extraction and representation capability of the model, leading to more robust and accurate LAI and LNC estimations.

C. Model Performance of Comparative Experiments

This study used 3-D CNN and CNN-LSTM to develop an LAI and LNC estimation model for individual *Populus* trees. Table V shows the performance of each LAI estimation model based on a 3-D CNN and CNN-LSTM architecture. According to the results, using selected bands as input for the 3-D CNN model has achieved the highest LAI and LNC estimation accuracy in each Scale. As shown in Fig. 18, Scale 2 achieved the best LAI estimation results with an R^2 of 0.829 $\pm$ 0.012 and an RMSE of 0.198 $\pm$ 0.006. In contrast, the best LNC estimation result was achieved in Scale 1, with an R^2 of 0.809 $\pm$ 0.007 and an RMSE of 1.358 $\pm$ 0.038 g/kg. Compared with the results of dual-branch models, the RMSE has increased by 9.39% and 10.41% in the LAI and LNC estimation models, respectively. Table V also shows the performance of each LNC estimation model based on the CNN-LSTM architecture. According to the results, using selected bands achieved the highest LAI and LNC estimation accuracy in each Scale. As shown in Fig. 18, the best LAI model was achieved in Scale 2 with an R^2 of 0.822 $\pm$ 0.011 and an RMSE of 0.203 $\pm$ 0.009, and the best LNC model was achieved in Scale 1 with an R^2 of 0.798 $\pm$ 0.010 and an RMSE of 1.388 $\pm$ 0.034 g/kg. Compared with the results of dual-branch models, the RMSE has increased by 12.15% and 12.85% in the LAI and LNC estimation models, respectively.

TABLE V

MEAN AND STANDARD DEVIATION OF R^2 VALUES OF LAI AND LNC ESTIMATION MODELS DEVELOPED USING 3-D CNN AND CNN-LSTM ARCHITECTURES

Model	Inputs	Scale 1	Scale 2	Scale 3	Scale 4
LAI estimation model based on 3-D CNN architecture	All bands	0.798 ± 0.014	0.814 ± 0.012	0.795 ± 0.013	0.770 ± 0.016
	Selected bands	0.818 ± 0.007	0.829 ± 0.012	0.807 ± 0.012	0.781 ± 0.018
	VIs	0.795 ± 0.007	0.809 ± 0.005	0.801 ± 0.005	0.779 ± 0.007
LNC estimation model based on 3-D CNN architecture	All bands	0.801 ± 0.007	0.758 ± 0.010	0.734 ± 0.011	0.717 ± 0.014
	Selected bands	0.809 ± 0.007	0.773 ± 0.009	0.749 ± 0.008	0.735 ± 0.010
	VIs	0.798 ± 0.006	0.766 ± 0.007	0.726 ± 0.013	0.704 ± 0.014
LAI estimation model based on CNN-LSTM architecture	All bands	0.804 ± 0.012	0.809 ± 0.013	0.798 ± 0.011	0.756 ± 0.021
	Selected bands	0.812 ± 0.009	0.822 ± 0.011	0.802 ± 0.018	0.758 ± 0.019
	VIs	0.790 ± 0.004	0.808 ± 0.015	0.793 ± 0.014	0.746 ± 0.021
LNC estimation model based on CNN-LSTM architecture	All bands	0.788 ± 0.012	0.760 ± 0.016	0.722 ± 0.025	0.707 ± 0.024
	Selected bands	0.798 ± 0.010	0.776 ± 0.014	0.740 ± 0.014	0.719 ± 0.025
	VIs	0.780 ± 0.014	0.762 ± 0.016	0.718 ± 0.026	0.699 ± 0.027

TABLE VI

COMPUTATIONAL EFFICIENCY COMPARISON OF THREE DL ARCHITECTURES

Phenotypic parameters	Model	Params (M)	Training time (min)	Inference time (min)
LAI	CNN–LSTM	5.97	67	18
	3D-CNN	8.45	148	61
	Dual-branch	7.64	95	34
LNC	CNN–LSTM	6.57	98	27
	3D-CNN	10.78	244	71
	Dual-branch	8.09	123	51

D. Comparative Experiment on Model Computational Efficiency and Robustness to Noise

Table VI shows the computational efficiency comparison of three DL architecture. The inference time refers to the time taken for the estimation of 2069 individual trees. For LAI estimation, the CNN–LSTM model exhibited the lowest computational cost, with 5.97 M trainable parameters, requiring 67 min for training and 18 min for inference. In contrast, the 3-D CNN showed the highest computational burden, with 8.45 M parameters, a training time of 148 min, and an inference time of 61 min. The dual-branch architecture achieved an intermediate level of computational complexity, containing 7.64 M parameters, with training and inference times of 95 and

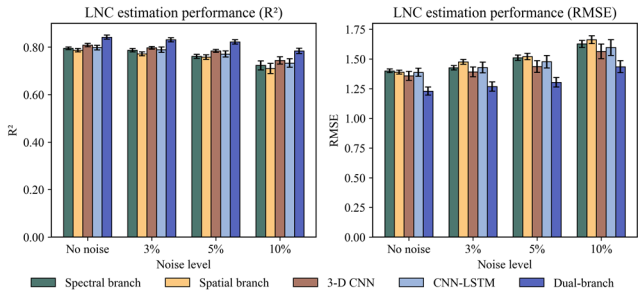


Fig. 20. LNC estimation accuracy of three hybrid architectures and two branch-based models under different noise levels.

34 min, respectively. For LNC estimation, the overall training and inference time costs increased across all models due to the higher spatial resolution of the input data. The CNN–LSTM, 3-D CNN, and dual-branch models contained 6.57, 10.78, and 8.09 M parameters, respectively, following a consistent ranking with that observed for LAI estimation. These results indicate that the computational cost of the dual-branch model lies between that of the CNN–LSTM and the 3-D CNN, while maintaining a more compact structure than the 3-D CNN.

Fig. 19 shows the LAI estimation accuracy of three hybrid architectures and two single-branch models under different noise levels. The single-branch models showed the greatest sensitivity to noise, with the spectral branch R^2 decreasing from 0.820 ± 0.006 to 0.741 ± 0.014 and RMSE increasing from 0.203 ± 0.011 to 0.245 ± 0.020, and the spatial branch R^2 decreasing from 0.817 ± 0.007 to 0.758 ± 0.012 and RMSE increasing from 0.202 ± 0.009 to 0.236 ± 0.017. The 3-D CNN achieved relatively higher accuracy under low-noise conditions but experienced a noticeable decline at higher noise levels, with R^2 dropping to 0.776 ± 0.021 and RMSE increasing to 0.227 ± 0.016. The CNN-LSTM model exhibited moderate robustness at low noise levels but showed a more pronounced performance degradation under strong noise perturbations, with R^2 decreasing from 0.822 ± 0.011 to 0.752 ± 0.018 and RMSE increasing from 0.203 to 0.239. In contrast, the dual-branch model consistently outperformed all other models across all noise levels, maintaining the highest R^2 values (0.861 ± 0.006–0.803 ± 0.011) and the lowest RMSE values (0.181 ± 0.007–0.213 ± 0.012), along with smaller standard deviations, indicating superior robustness and stability under noisy conditions.

Fig. 20 shows the LNC estimation accuracy of three hybrid architectures and two single-branch models under different noise levels. The results are similar as the LAI estimations. The single-branch models exhibited the highest sensitivity to noise, with the spectral branch R^2 decreasing from 0.795 ± 0.006 to 0.723 ± 0.019, while the spatial branch showed a similar trend, with R^2 declining from 0.787 ± 0.007 to 0.710 ± 0.022. The 3-D CNN model achieved higher estimation accuracy than CNN-LSTM in each noise level. The 3-D CNN achieved relatively higher accuracy under low-noise conditions but experienced a noticeable performance degradation as noise increased, with R2 decreasing to 0.744 ± 0.015 and RMSE increasing to 1.564 ± 0.061 g/kg at the 10% noise level. The CNN–LSTM model demonstrated moderate robustness under weak noise perturbations but showed a more pronounced decline under strong noise, with R^2 decreasing from 0.798 ± 0.010 to 0.733 ± 0.018 and RMSE increasing from 1.388 ± 0.034 g/kg to 1.596 ± 0.066 g/kg. In contrast, the dual-branch model consistently outperformed all other models across all noise levels, maintaining the highest R^2 values (0.842 ± 0.009 to 0.784 ± 0.011) and the lowest RMSE values (1.230 ± 0.034 g/kg to 1.435 ± 0.050 g/kg), together with relatively smaller standard deviations, indicating superior robustness and stability for LNC estimation under noisy conditions.

IV. Discussion

A. Influence of Spatial Resolution on LAI and LNC Estimation for Individual Populus Trees

Determining the appropriate spatial resolution is crucial for accurately estimating LAI and LNC at the individual tree scale using UAV-based remote sensing technology, as it directly influences the ability to distinguish fine structural details and biochemical properties. An appropriate spatial resolution ensures both precision and efficiency in capturing tree-level variations. Previous studies have investigated the impact of the spatial resolution of UAV multispectral imagery on the accuracy of LAI estimation for certain crops, such as sugar beet and rapeseed. Findings suggest that estimation accuracy declines significantly when spatial resolution decreases beyond a specific threshold [17], [18]. This threshold can thus inform practical applications by balancing inversion precision with data acquisition efficiency. In this study, the optimal spatial resolution for estimating the LAI of individual *Populus* trees was determined to be Scale 2 (0.10 m/pixel). When the spatial resolution was reduced below 0.10 m/pixel, a gradual decline in R^2 was observed. This result is consistent with previous findings. Notably, the accuracy obtained using Scale 1 was slightly lower than that of Scale 2, indicating that ultrahigh spatial resolution does not necessarily yield better estimation results in this context. This may be attributed to the fact that this study focused on individual *Populus* trees, which exhibit characteristics such as sparse foliage and heterogeneity compared to crops. Consequently, the Scale 1 (0.05 m/pixel) data captured fine details, including intercrown gaps, shadows, and branches, which might be misinterpreted as lower vegetation density during feature extraction, reducing LAI estimation accuracy. In contrast, the Scale 2 (0.10 m/pixel) data averaged out some local noise, better representing the overall canopy characteristics and yielding more stable LAI estimation results. Chakhvashvili et al. [70] estimated the LAI and LCC under changing spatial scale using UAV multispectral data; the results indicate that the inversion of biochemical parameters such as LCC benefits from high spatial resolution, with greater improvement observed compared to the inversion of LAI. These findings are consistent with the results of this study. For instance, Figs. 11 and 14 illustrate that the estimation accuracy of LNC increases with higher spatial resolution. In particular, the results at Scale 1 are significantly better than those at other scales. The greater sensitivity of

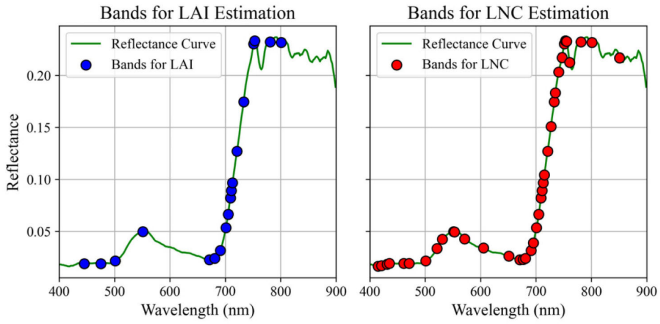


Fig. 21. Bands used in VIs for LAI and LNC estimation.

LNC estimation to spatial resolution, compared to LAI, may be primarily due to differences in their biophysical and spectral characteristics. LNC is a leaf-level biochemical trait that relies heavily on subtle spectral variations, particularly in the RE and NIR regions, which are easily affected by background noise such as soil and shadows. High spatial resolution enables more accurate isolation of leaf-dominated pixels, minimizing the influence of nonvegetative components and improving the fidelity of spectral signals related to nitrogen content. In contrast, LAI represents a canopy-scale structural parameter with lower spatial variability, and its estimation primarily depends on the overall vegetation cover, which can be sufficiently captured even at coarser resolutions. As a result, increasing spatial resolution has a more pronounced impact on the accuracy of LNC estimation than on LAI.

B. Influence of the Feature Form on LAI and LNC Estimation for Individual Populus Trees

This study explored the influence of different feature forms on estimating LAI and LNC for individual *Populus* trees. The comparison of these feature forms provides valuable insights into their respective strengths and limitations in capturing the essential information for accurate biophysical and biochemical parameter estimation. At Scales 3 and 4, full-band spectral data demonstrated superior LAI and LNC estimation performance compared to feature-selected bands and VIs. This advantage can be attributed to the comprehensive spectral information captured by full-band data, compensating for the increased background noise and mixed pixel effects characteristic of lower resolution imagery. However, as spatial resolution improves, the finer details of the tree structure become more discernible. In this case, feature-selected bands and VIs, which focus on specific wavelengths correlated with LAI and LNC, tend to perform better at Scales 2 and 1. The higher resolution data allows for more precise targeting of spectral bands sensitive to specific physiological characteristics. Therefore, using feature-selected bands or VIs is more effective for constructing estimation models, while using all bands introduces noise from irrelevant bands and increases the risk of overfitting.

In the dual-branch estimation model, the selection of feature forms is related to the branch type. Fig. 21 illustrates the spectral bands used in the VIs. Compared to the band selection results, the VIs employed for LAI modeling lack bands in the 500–600 nm range and the near-infrared region. In contrast, the VIs used for LNC modeling cover the relevant sensitive spectral ranges. As shown in Tables II and III, within the spectral branches at the fine Scale (Scale 1 and Scale 2), the performance of the selected bands consistently surpasses that of the VIs. This result indicates that, regardless of whether the VIs contain sufficient spectral information, their performance in the spectral branch remains inferior to that of the selected bands. Two main factors can explain the observed results. First, excluding nonvegetative pixels in the spectral branch leads to a purer spectral signal, allowing the selected bands to characterize the target variables more effectively. Second, the LSTM algorithm in the spectral branch captures dependencies between bands. In contrast, VIs are treated as independent features, which limits their ability to represent spectral relationships and results in lower performance. Within the spectral branches at the fine scale (Scale 1 and Scale 2), the performance of the VIs consistently surpasses that of the selected bands. This is because VIs integrate spectral information from multiple bands, which enables them to capture complex canopy structures and vegetation distribution patterns. In particular, when the data contains nonvegetative pixels, VIs can effectively reduce background noise and enhance the extraction of vegetation-relevant information. The observed superiority of VIs may also be attributed to the stronger compatibility between VIs and the structural characteristics of the model. The shallow 2-D CNN is designed to capture low-level spatial features such as texture and structure, which are more clearly expressed in VIs due to their enhanced spatial consistency and contrast. Moreover, using a residual learning module and SAM further strengthens the advantage of VIs. These modules benefit from inputs that highlight important regions and suppress noise, a characteristic inherently possessed by VIs.

C. Advantages of Dual-Branch DL Architecture

This study developed a dual-branch DL architecture for LAI and LNC estimation of individual *Populus* trees. The dual-branch architecture demonstrated superior estimation accuracy compared to results obtained using single-branch models, 3-D CNN, and LSTM. This improvement can be attributed to several factors. First, the dual-branch architecture can simultaneously process different data types, enabling the model to exploit the complementary nature of multisource information fully. Comparative experiment shows that the spectral and spatial branches achieve moderate accuracy, whereas the full dual-branch model performs markedly better. Moreover, with increasing noise levels, single-branch models degrade rapidly, whereas the dual-branch model experiences a slower performance decline. This demonstrates that the improvement stems from the integration of complementary information rather than from an increase in the number of model parameters. Second, although the 3-D CNN and CNN-LSTM architectures can extract both spectral and spatial features, they often handle a single data input with one scale and one feature form, which restricts the learning capacity of DL models. In contrast, the dual-branch architecture can process different types of data separately, which enables a more comprehensive capture of features across different dimensions. For instance, using

VIs as input for 3-D CNN and CNN-LSTM models often achieves lower estimation accuracy, since both models are adept at handling sequential data such as time-series and hyperspectral data. In contrast, the results of the spatial branch indicated that using VIs as input can improve the estimation accuracy of LAI. Moreover, each branch can optimize its feature extraction approach based on the type of data, which helps prevent information loss during the feature extraction process. Although 3-D CNN and CNN-LSTM models handle spatial and spectral data simultaneously, they typically use a unified feature extraction method for all data types. This may lead to excessive fusion or neglect of certain features.

D. Limitations and Future Works

While the findings of this study represent significant progress, several limitations and opportunities for further improvement remain. First, as the application scenario of this study focuses on the cultivation of superior tree varieties in relatively small-scale areas, the proposed model may face challenges when extended to large-area estimation of LAI and LNC for Populus plantations. Future work will therefore explore the integration of the dual-branch DL model with 3-D radiative transfer models (e.g., DART and LESS) to enhance its transferability across different spatial scales. In addition, BRDF effects pose a notable challenge for the quantitative analysis and spectral interpretation of remotely sensed data, particularly for high-resolution UAV-based hyperspectral observations. In this study, the experimental area was relatively limited, resulting in an effective flight duration of approximately one hour. Moreover, hyperspectral data were acquired using a consistent flight direction, high image overlap, and near-nadir viewing geometry, which helped mitigate BRDF-related effects. Nevertheless, under large-scale application scenarios with extended acquisition durations, BRDF effects may become more pronounced and should not be neglected. Accordingly, future studies will focus on collecting multiangle UAV hyperspectral data and implementing BRDF correction strategies to further reduce angular effects and improve the accuracy of LAI and LNC estimation. Furthermore, due to the experimental design of the study area, each sampled *Populus* tree corresponds to a unique genotype, which prevents the implementation of genotype-stratified validation strategies in the current study. Future research will aim to identify or establish suitable experimental sites where multiple individuals share the same genotype, thereby enabling more rigorous genotype-aware evaluation schemes, such as leave-one-genotype-out cross-validation. Finally, this study utilized LiDAR data solely for individual tree segmentation, which limits the exploitation of 3-D structural information. Future work will further explore the fusion of hyperspectral and LiDAR data to fully leverage complementary spectral and structural features for more accurate estimation of biophysical and biochemical parameters.

V. Conclusion

This study developed a dual-branch DL architecture for estimating the LAI and LNC of individual *Populus* trees. This architecture has significantly improved estimation accuracy compared to traditional DL methods such as 3-D CNN and CNN-LSTM. Based on the dual-branch DL architecture, the LAI and LNC estimation model achieved an R^2 of 0.861 ± 0.006 and 0.842 ± 0.009, respectively. The main conclusions of this study are as follows.

1) The spatial scale of hyperspectral imagery influences the estimation results of LAI and LNC. The increasing spatial resolution has a more pronounced impact on the accuracy of LNC estimation than on LAI. In this study, the optimal scale for LAI and LNC estimation is Scale 2 (0.10 m/pixel) and Scale 1 (0.05 m/pixel), respectively.
2) When applying the DL algorithm for LAI and LNC estimation of individual *Populus* tree, selected bands and VIs are more effective for LAI and LNC estimation at finer scales (Scales 1 and 2), whereas all band data provides superior accuracy at lower spatial resolutions.
3) At Scales 1 and 2, selected bands enhance the accuracy of the spectral branch, while VIs are more effective in improving the accuracy of the spatial branch.
4) The proposed architecture can extract deep features from different feature forms (selected bands and VIs) and different spatial scales (Scales 1 and 2), contributing to more accurate LAI and LNC estimation results for individual *Populus* trees.

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